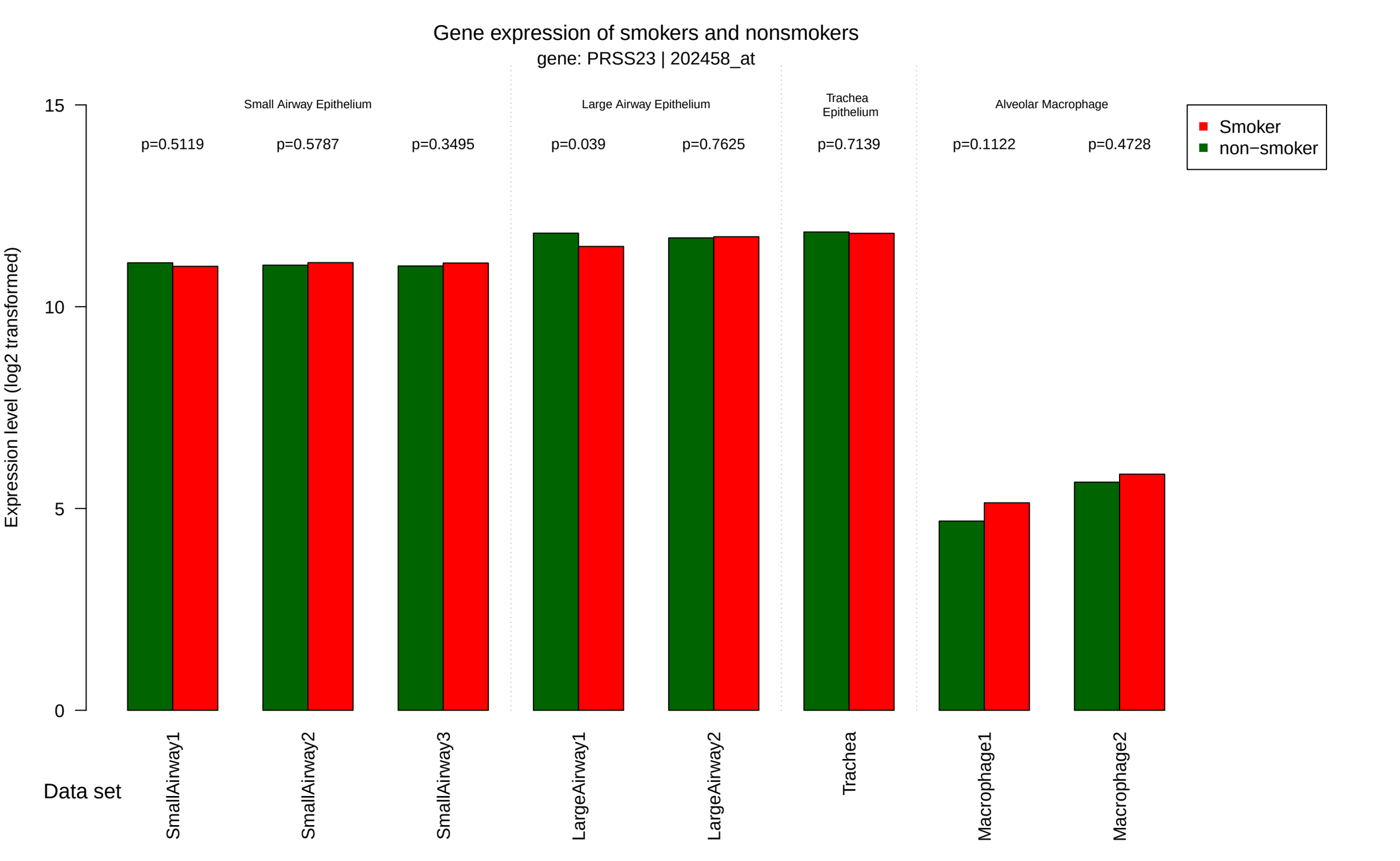
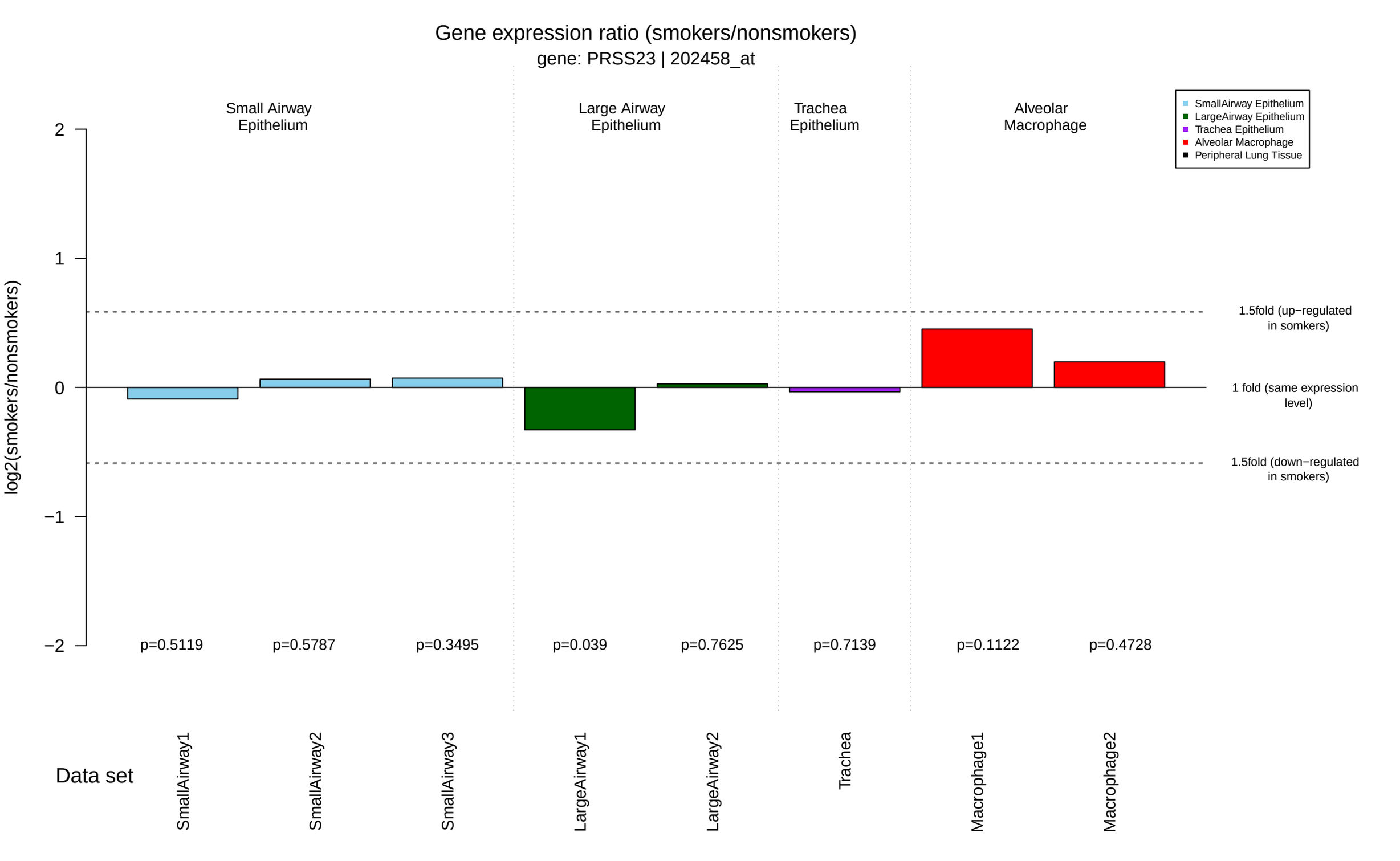
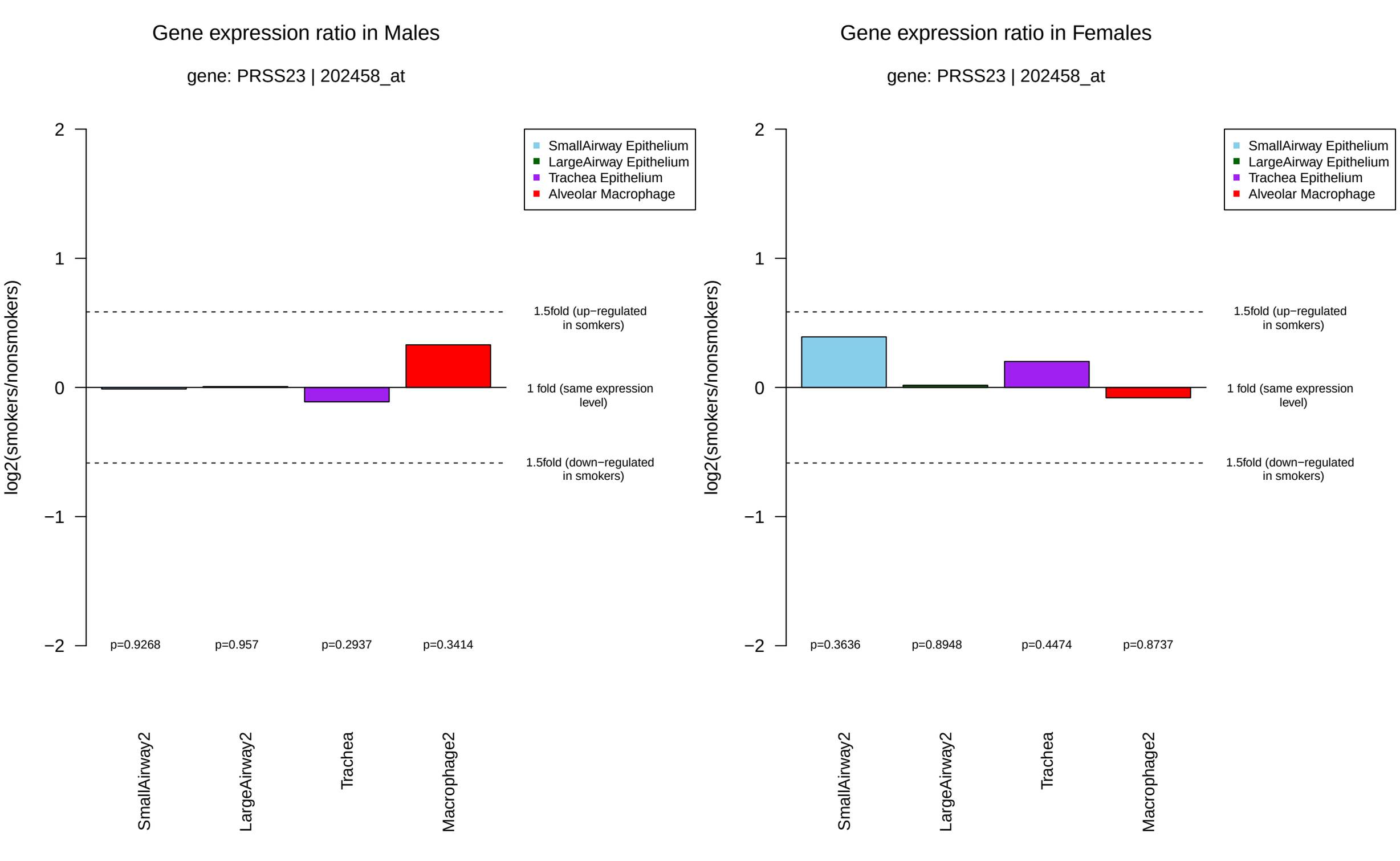
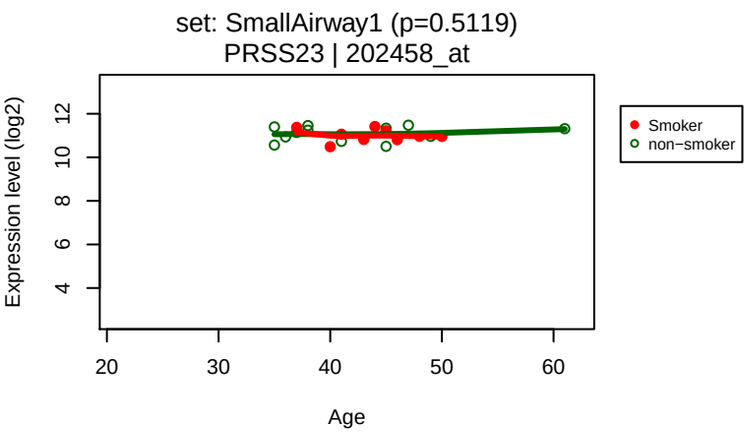
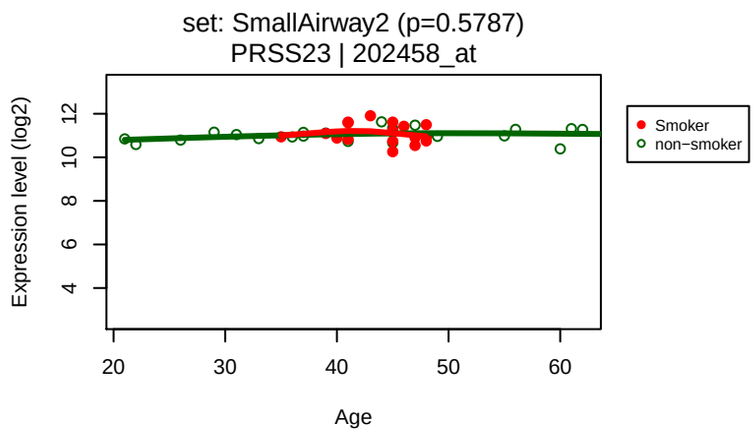
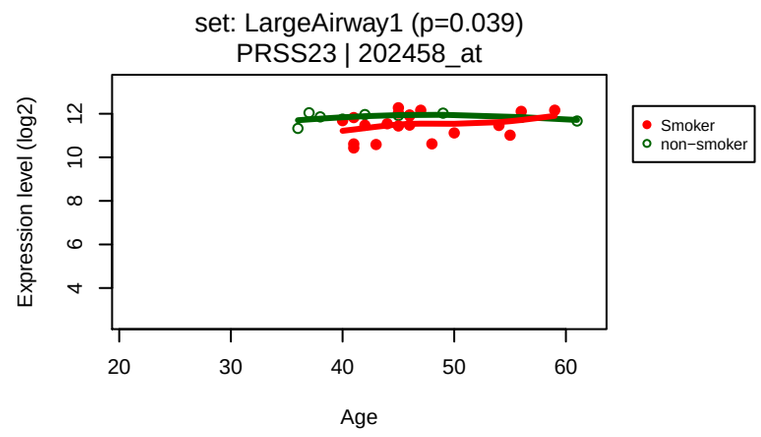

Gene expression of smokers and nonsmokers
gene: PRSS23 | 202458_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5119
p=0.5787
p=0.3495
p=0.039
p=0.7625
p=0.7139
p=0.1122
p=0.4728
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

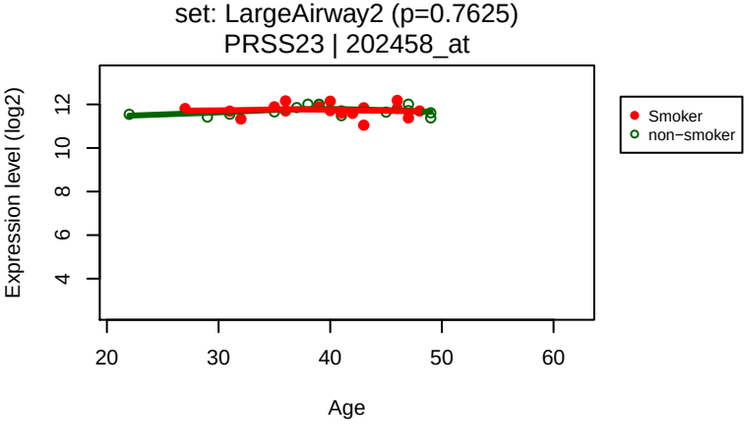

Gene expression ratio (smokers/nonsmokers)
gene: PRSS23 | 202458_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5119
p=0.5787
p=0.3495
p=0.039
p=0.7625
p=0.7139
p=0.1122
p=0.4728
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

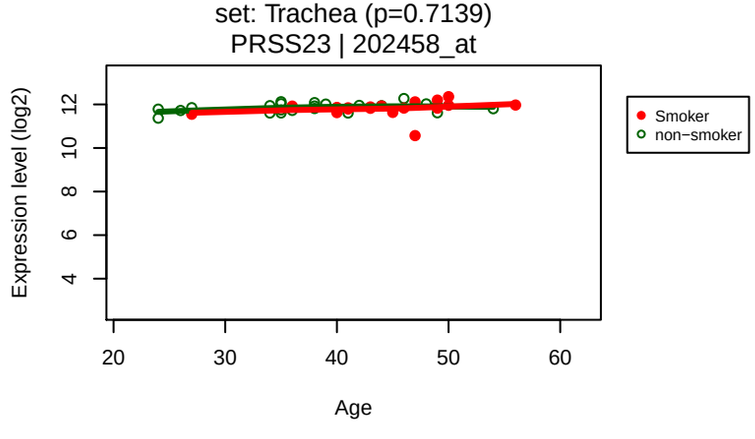

Gene expression ratio in Males
gene: PRSS23 | 202458_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9268
p=0.957
p=0.2937
p=0.3414
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS23 | 202458_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3636
p=0.8948
p=0.4474
p=0.8737
SmallAirway2
LargeAirway2
Trachea
Macrophage2

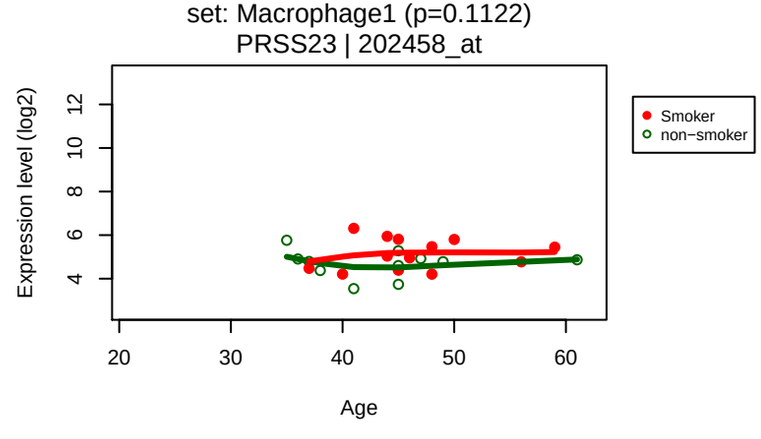

set: SmallAirway1 (p=0.5119)
PRSS23 | 202458_at
Expression level (log2)
Age
Smoker
non−smoker

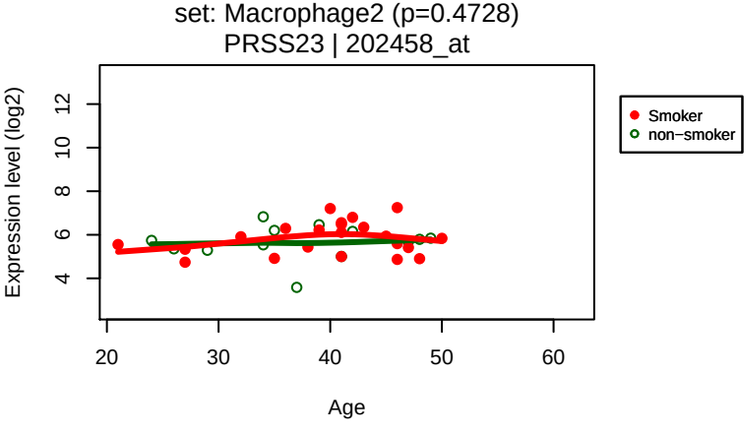

set: SmallAirway2 (p=0.5787)
PRSS23 | 202458_at
Expression level (log2)
Age
Smoker
non−smoker

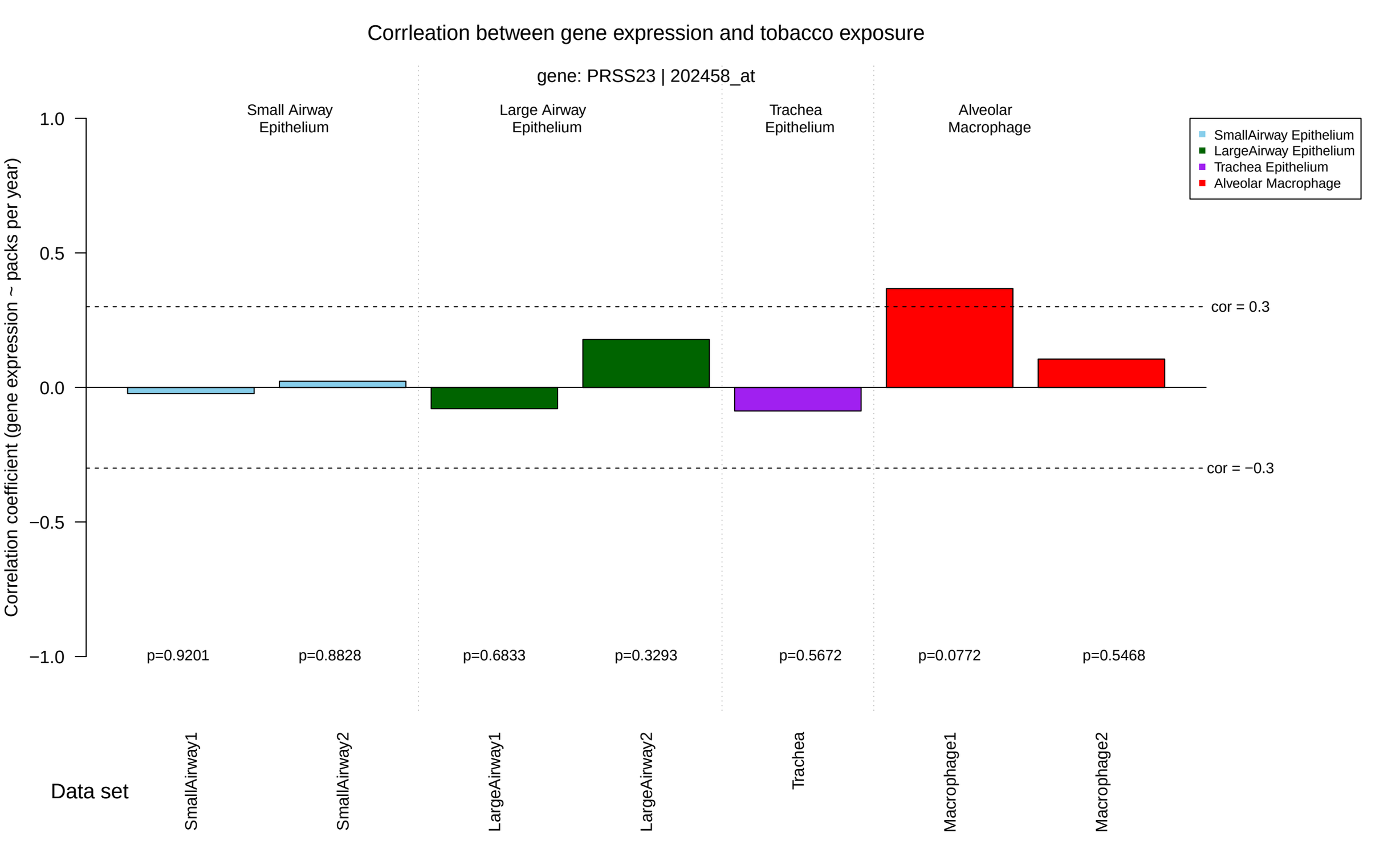

set: LargeAirway1 (p=0.039)
PRSS23 | 202458_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7625)
PRSS23 | 202458_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7139)
PRSS23 | 202458_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1122)
PRSS23 | 202458_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4728)
PRSS23 | 202458_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS23 | 202458_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9201
p=0.8828
p=0.6833
p=0.3293
p=0.5672
p=0.0772
p=0.5468
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2